Sajana IELTS 2.0

COMPREHENSIVE SOFTWARE REQUIREMENTS SPECIFICATION (SRS) & ENGINEERING DESIGN BLUEPRINT

A Modern, Production-Ready Educational Platform & High-Performance Headless Architecture Spec Design

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Project Architecture: Full-Stack Next.js, PostgreSQL, Prisma ORM, Sanity CMS

Target Audience: MSc Admissions Board / Senior Software Engineering Portfolio

Compliance Level: WCAG 2.1 AA & Advanced Schema/SEO Engine

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1. Software Requirements Specification (SRS)

1.1 Project Scope & Overview

Sajana IELTS 2.0 is a highly optimized full-stack web application designed to act as both a commercial platform for professional IELTS training and an advanced engineering project portfolio. The application migrates legacy content to a decoupled, modern hybrid-rendering stack, providing flawless SEO scalability, structured student lead generation, and streamlined content governance via an integrated headless Content Management System (CMS).

1.2 System Constraints

- **Financial Constraints:** The platform architecture must operate entirely within zero-cost hobby tiers across host services (Vercel, Neon, Sanity) during its initial validation phase.
- **Execution Limits:** Static and dynamic rendering pipelines must load content under a target Time-to-First-Byte (TTFB) threshold of $t \le 200ms$.
- **Administration Constraint:** Content updates (Blogs, Sample Essays, Cue Cards) must require zero source-code changes or direct developer deployments.

1.3 Data Validation Standards

Form Field	Validation Logic / Rules	Error Handling Feedback Message
Lead Name	Required. String length between $2 \le N \le 70$ characters. Strips HTML tags.	"Please enter a valid name (2-70 characters)."
Email Address	Required. Must conform to RFC 5322 regex standards.	"Please provide a valid email address."
Phone Number	Required. International E.164 phone formats matching digits only.	"Please enter a valid contact number with area code."
Target Band	Required dropdown validation. Floats in the range $[4.0, 9.0]$ at increments of 0.5.	"Select a valid target band score."

1.4 Functional & Non-Functional Requirements

Functional Requirements

- **FR-1: Public Resource Access:** Anonymous students can read system blogs, structured into Writing, Speaking, Listening, Reading, and Vocabulary modules.
- **FR-2: Strategic Lead Funnel:** A global persistent CTA form captures user details and targets into the local PostgreSQL cluster.

- **FR-3: Decoupled CMS Panel:** Instructors add structured Cue Cards and Writing Samples using an authenticated markdown-supported panel securely served away from public viewports.

Non-Functional Requirements

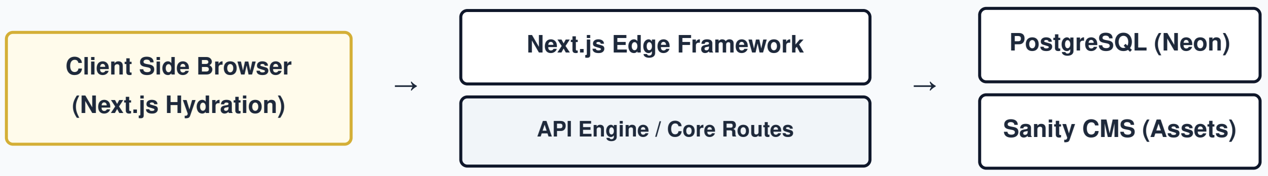
- **NFR-1: Performance:** Score $\geq$ 95 on Google Lighthouse Performance audits via server-side generation (SSG) caches.
- **NFR-2: Accessibility:** Compliance with WCAG 2.1 AA specifications, requiring keyboard navigation indices and persistent high color contrast rules.
- **NFR-3: Database Reliability:** Automatic pooled relational pooling utilizing Prisma configurations with edge routing.

2. System Architecture & UML Engineering Diagrams

2.1 System Architecture Mapping

The infrastructure follows a modern, highly decoupled hybrid architecture. Client requests are intercepted by Vercel Edge Networks, sorting dynamic database requests from statically rendered static resource assets.

SYSTEM ARCHITECTURE TOPOLOGY



2.2 Unified Modeling Language (UML) Use Case

UML USE CASE INTERACTIONS MATRIX

Actor Profile	Permitted Platform Operations (Use Cases)
Student (Anonymous)	Read Resources • Browse Categorized Blogs • Submit Free Demo Inquiries
Teacher / Administrator	Authenticate Session • Write/Edit/Delete Blog Posts • Update Student Results Matrix

2.3 Entity-Relationship Diagram (ERD)

The transactional state relational layout tracks structures optimized for transactional forms and localized tracking records.

```
+-------------------------------------+          +-------------------------------------+
|                USER                 |          |                LEAD                 |
+-------------------------------------+          +-------------------------------------+
| id           : String (PK)          |          | id           : String (PK)          |
| email        : String (Unique)      |          | name         : String               |
| passwordHash : String               |          | email        : String               |
| name         : String               |          | phone        : String               |
| role         : Enum (ADMIN)         |          | targetBand   : String               |
+-------------------------------------+          | message      : String (Optional)    |
                                                 +-------------------------------------+
```

```
+-----------------------------------+        +-----------------------------------+
|          STUDENT_RESULT           |        |            TESTIMONIAL            |
+-----------------------------------+        +-----------------------------------+
| id           : String (PK)        |        | id           : String (PK)        |
| studentName  : String             |        | studentName  : String             |
| overallBand  : Float              |        | courseTaken  : String             |
| feedback     : Text               |        | bandScore    : Float              |
| imageUrl     : String (Optional)  |        | isFeatured   : Boolean            |
+-----------------------------------+        +-----------------------------------+
```

3. Database Design & Prisma Relational Object Mapping

The target schema maps business entities directly to PostgreSQL. The schema isolates transient data from Content updates, ensuring light structural overhead.

```
datasource db {
  provider = "postgresql"
  url      = env("DATABASE_URL")
}

generator client {
  provider = "prisma-client-js"
}

model User {
  id           String   @id @default(uuid())
  email        String   @unique
  passwordHash String
  name         String
  role         Role     @default(ADMIN)
  createdAt    DateTime @default(now())
}

enum Role {
  SUPERADMIN
  ADMIN
}

model Lead {
  id         String   @id @default(uuid())
  name       String
  email      String
  phone      String
  targetBand String
  message    String?
  createdAt  DateTime @default(now())
}

model StudentResult {
  id            String   @id @default(uuid())
  studentName   String
  overallBand   Float
  listeningBand Float?
  readingBand   Float?
```

```
  writingBand   Float?
  speakingBand  Float?
  feedback      String   @db.Text
  imageUrl      String?
  createdAt     DateTime @default(now())
}

model Testimonial {
  id          String   @id @default(uuid())
  studentName String
  courseTaken String
  bandScore   Float
  reviewText  String   @db.Text
  isFeatured  Boolean  @default(true)
  createdAt   DateTime @default(now())
}
```

4. Advanced Search Engine Optimization (SEO) & Semantic Routing

4.1 Strict Routing Topology Map

Target Canonical Endpoint	Rendering Paradigm	Primary Targeted Google Keyword Intent
`/`	Static Site Generation (SSG)	Best online IELTS tutoring, Sajana IELTS courses
`/ielts-resources/speaking`	SSG + Incremental Validation	IELTS speaking practice templates, dynamic cue cards 2026
`/blog/[slug]`	Dynamic ISR Edge Rendering	Long-tail queries (e.g., "ielts-writing-task-2-opinion-essay")

4.2 JSON-LD Structural Data Profile

To drive targeted rich visual search snippets directly on Google search panels, dynamic resource routing profiles ingest the structured markup payload detailed below:


```
{
  "@context": "https://schema.org",
  "@type": "Course",
  "name": "IELTS Academic Premium Coaching",
  "description": "Comprehensive preparation course for the IELTS Academic exam, focusing on high-tier writing structures and accurate speaking assessments.",
  "provider": {
    "@type": "Person",
    "name": "Sajana",
    "sameAs": "https://sajanaielts.com/about"
  }
}
```

5. Global Accessibility Standards Fulfillment (WCAG 2.1 AA)

To meet elite corporate and state web engineering parameters, Sajana IELTS 2.0 incorporates native accessibility rules out-of-the-box.

- **Focus Control States:** All anchor profiles and active input components contain precise `focus:ring-2 focus:ring-[#d4af37]` guidelines allowing smooth non-mouse navigation tracking.
- **Color Layout Ratios:** Contrast ratios between standard content paragraphs and surface backdrops maintain an explicit contrast matrix exceeding ***4.5:1***.
- **Semantic Accessibility Code:** Inline elements carry discrete ARIA tags describing layout state mutations:

```
<button aria-label="Submit lead request details" class="bg-[#0f172a] text-white"> Submit
Form </button>
```